The system combines geographic data including street names, points of interest, railroads and airports with live video from the camera system. The system offers images in picture mode that allow it to evoke a general view of the arena that is surrounding the camera's field-of-view.

Alive app by Times of India. Alive is an augmented reality based application used to view extensive content like animation, video and photo shoots. You can scan a printed image in newspapers, brochures, pamphlets or magazines, and the app recognises images, logos, location, QR codes and objects in real time.

How augmented reality works

This technology works on the basis of two types of approaches: marker based and location based.

Marker based augmented reality. The QR code system is one of the most common methods of marker based augmented reality. It consists of black squares arranged in a square grid on a white background, which can be easily read by an imaging device such as a camera. For this to work, you need a smartphone that has a marker based augmented reality application like QR Code that can scan a pattern such as a barcode or symbol through the camera of the mobile.

The software recognises the pattern and superimposes a digital image on your mobile screen, and gives you a 3D or an animated digital image for a better experience as compared to a 2D image.

Location based augmented reality. With a smartphone and a location based application system like GPS, you can point the augmented reality based application towards a real scene. GPS software recognises the device location in the application. Based on the recorded location and orientation by the inbuilt sensor of your phone, you can predict the overview data of the relevant location. Digital informative data is matched to the real scene, which is then visible through the device camera.

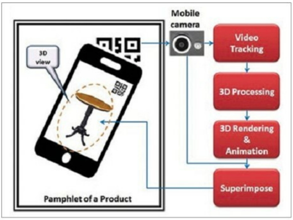

Fig. 4: Marker based augmented reality application

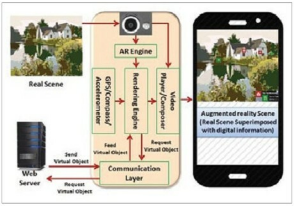

Fig. 5: Location based augmented reality application

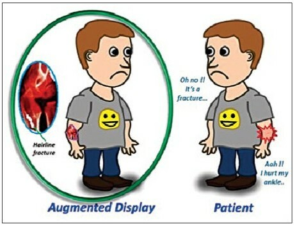

Fig. 6: Augmented reality as superimposition

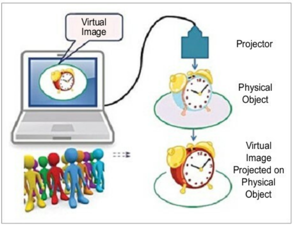

Fig. 7: Augmented reality as projection

One such application of the location based augmented reality is navigation. It enhances the effectiveness of augmented navigation devices. Information is displayed on the automobile's windshield and the system indicates the direction and distance of the destination along with weather and terrain conditions, traffic information and potential hazards.

Superimposition based augmented reality. In this case, the augmented view is superimposed on the real view of an object. Augmented view is either the scanned image of the real object or the internal view of the object. Applications of this kind of augmented reality are in education, defence, medical and architecture, among others.

Projection based augmented reality. With the help of this process, you can project virtual images onto physical objects. This method of having a projection of the virtual image is known as the projection augmented model, or PA model. You can even touch, feel and grasp the virtual image with your hands.

Future scope of augmented reality

Some areas where this technology is being developed are films, the retail sector (the technology is closing the gap between online and offline) and computer games.